AI-Driven Crop Recommendation: Optimising Indian Agriculture

Transforming traditional agricultural practices through precision soil analysis and machine learning. This project addresses the critical gap between generalized farming knowledge and localized soil conditions, leveraging data science to maximize crop yields and preserve soil health across diverse Indian agricultural landscapes.

Project Goals & Deliverables

Data Integration

Successfully process and clean real-world agricultural data containing nutrient levels and climate metrics from diverse Indian soil profiles.

Model Development

Train a highly accurate multi-class classification model capable of distinguishing between 22 different crop profiles with scientific precision.

System Evaluation

Rigorously test the model using standardized metrics to ensure reliability on unseen soil samples and real-world farming conditions.

Practical Application

Create an accessible framework where farmers can input raw soil test results and receive immediate, scientifically sound crop recommendations.

Data Acquisition & Preprocessing

Dataset Foundation

Utilizing a robust, open-source dataset hosted on Figshare, representing diverse Indian agricultural conditions from multiple soil testing laboratories.

Input Features

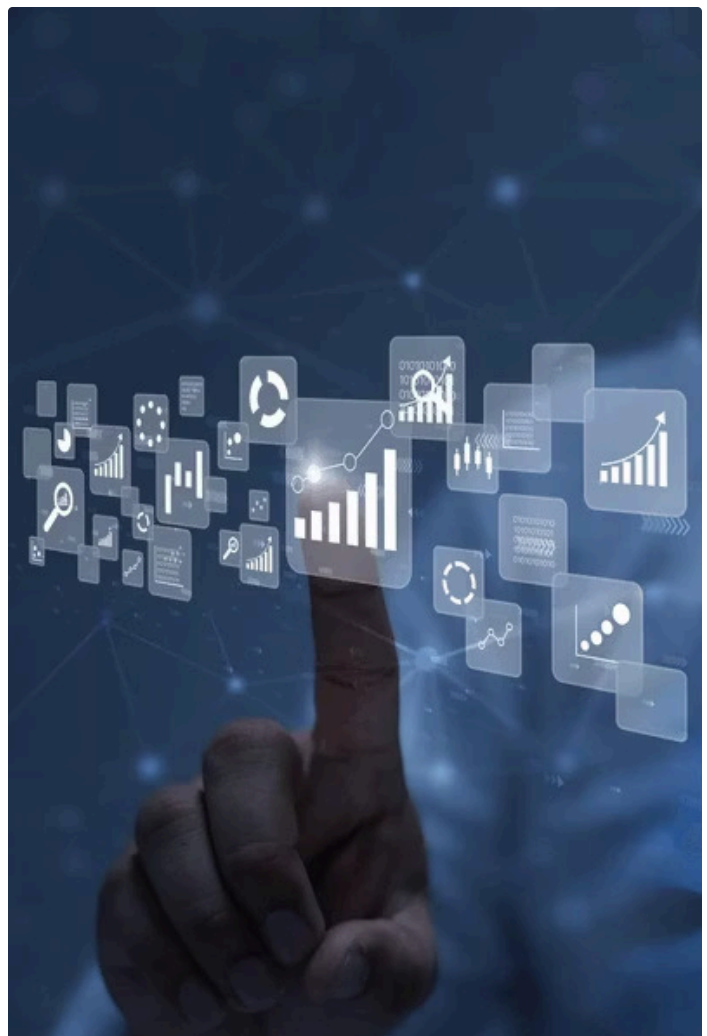
Analyzing seven key variables that determine crop suitability: Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, pH, and Rainfall measurements.

Target Variable

The dataset maps these environmental conditions to 22 distinct crop categories including Rice, Maize, Chickpea, Coffee, and other region-specific cultivars.

Preprocessing Strategy

Handling missing values through statistical imputation and preparing the data matrix for algorithms to ensure accurate scaling across different measurement units.



Machine Learning Pipeline

Data Splitting

Implementing an 80/20 train-test split to train the model and validate accuracy on hidden data representing real-world conditions.

Primary Algorithm

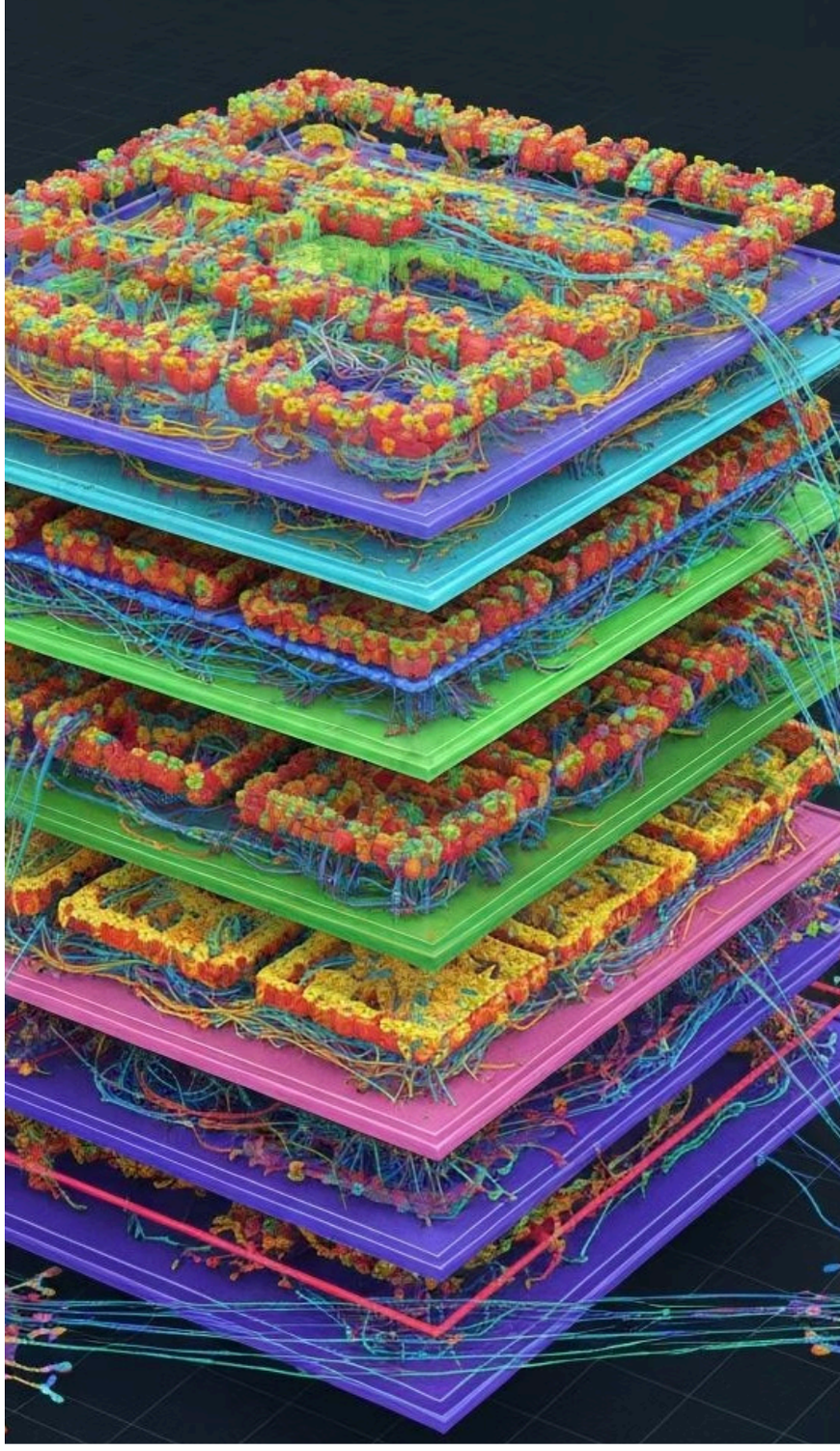
Deploying a Random Forests Classifier to establish clear, interpretable mathematical rules for crop selection that farmers can understand.

Comparative Models

Benchmarking against Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Decision Trees to guarantee highest accuracy.

Validation Metrics

Validating the final system using Accuracy, Precision, Recall, and F1-score to ensure reliability for agricultural decision-making.



Proposed Benefits & Future Value

Maximised Yields

Ensuring cultivators plant crops that their specific soil naturally supports, minimizing the risk of harvest failure and optimizing productivity.

Resource Efficiency

Reducing reliance on artificial fertilizers and excessive irrigation by working with the soil's existing chemical profile, lowering input costs.

Sustainable Farming

Preserving long-term soil health by encouraging data-backed crop rotation and preventing nutrient depletion across seasons.

Economic Empowerment

Providing rural and informal agricultural sectors with enterprise-grade, data-driven decision-making tools to increase profitability and livelihood security.

ITAPRactice Version control itaprojectcode Unlock Pro

Project main.py itaprojectcode.py ITACLASSASSIGNMENT.py

Run itaprojectcode

OUTPUT SCREENSHOTS

```
/usr/local/bin/python3.13 /Users/sidb/Desktop/ITAPRactice/itaprojectcode.py
Accuracy: 99.32%

Classification Report:
              precision    recall  f1-score   support

       apple       1.00      1.00      1.00        23
      banana       1.00      1.00      1.00        21
   blackgram       1.00      1.00      1.00        20
    chickpea       1.00      1.00      1.00        26
     coconut       1.00      1.00      1.00        27
      coffee       1.00      1.00      1.00        17
      cotton       1.00      1.00      1.00        17
      grapes       1.00      1.00      1.00        14
        jute       0.92      1.00      0.96        23
 kidneybeans       1.00      1.00      1.00        20
      lentil       0.92      1.00      0.96        11
       maize       1.00      1.00      1.00        21
       mango       1.00      1.00      1.00        19
   mothbeans       1.00      0.96      0.98        24
    mungbean       1.00      1.00      1.00        19
   muskmelon       1.00      1.00      1.00        17
      orange       1.00      1.00      1.00        14
      papaya       1.00      1.00      1.00        23
  pigeonpeas       1.00      1.00      1.00        23
 pomegranate       1.00      1.00      1.00        23
        rice       1.00      0.89      0.94        19
  watermelon       1.00      1.00      1.00        19

    accuracy                           0.99       440
   macro avg       0.99      0.99      0.99       440
weighted avg       0.99      0.99      0.99       440
```

```
   chickpea       1.00      1.00      1.00        26
    coconut       1.00      1.00      1.00        27
     coffee       1.00      1.00      1.00        17
     cotton       1.00      1.00      1.00        17
     grapes       1.00      1.00      1.00        14
       jute       0.92      1.00      0.96        23
kidneybeans       1.00      1.00      1.00        20
     lentil       0.92      1.00      0.96        11
      maize       1.00      1.00      1.00        21
      mango       1.00      1.00      1.00        19
  mothbeans       1.00      0.96      0.98        24
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  muskmelon       1.00      1.00      1.00        17
     orange       1.00      1.00      1.00        14
     papaya       1.00      1.00      1.00        23
  pigeonpeas       1.00      1.00      1.00        23
 pomegranate       1.00      1.00      1.00        23
        rice       1.00      0.89      0.94        19
  watermelon       1.00      1.00      1.00        19

    accuracy                           0.99       440
   macro avg       0.99      0.99      0.99       440
weighted avg       0.99      0.99      0.99       440

--- Crop Recommendations for New Soil Data ---
     N    P    K Recommended_Crop
0   90   42   43             rice
1   14   15   14           orange
2   60   55   44             rice
3  104   18   30           coffee
4    2  144  200           grapes

Process finished with exit code 0
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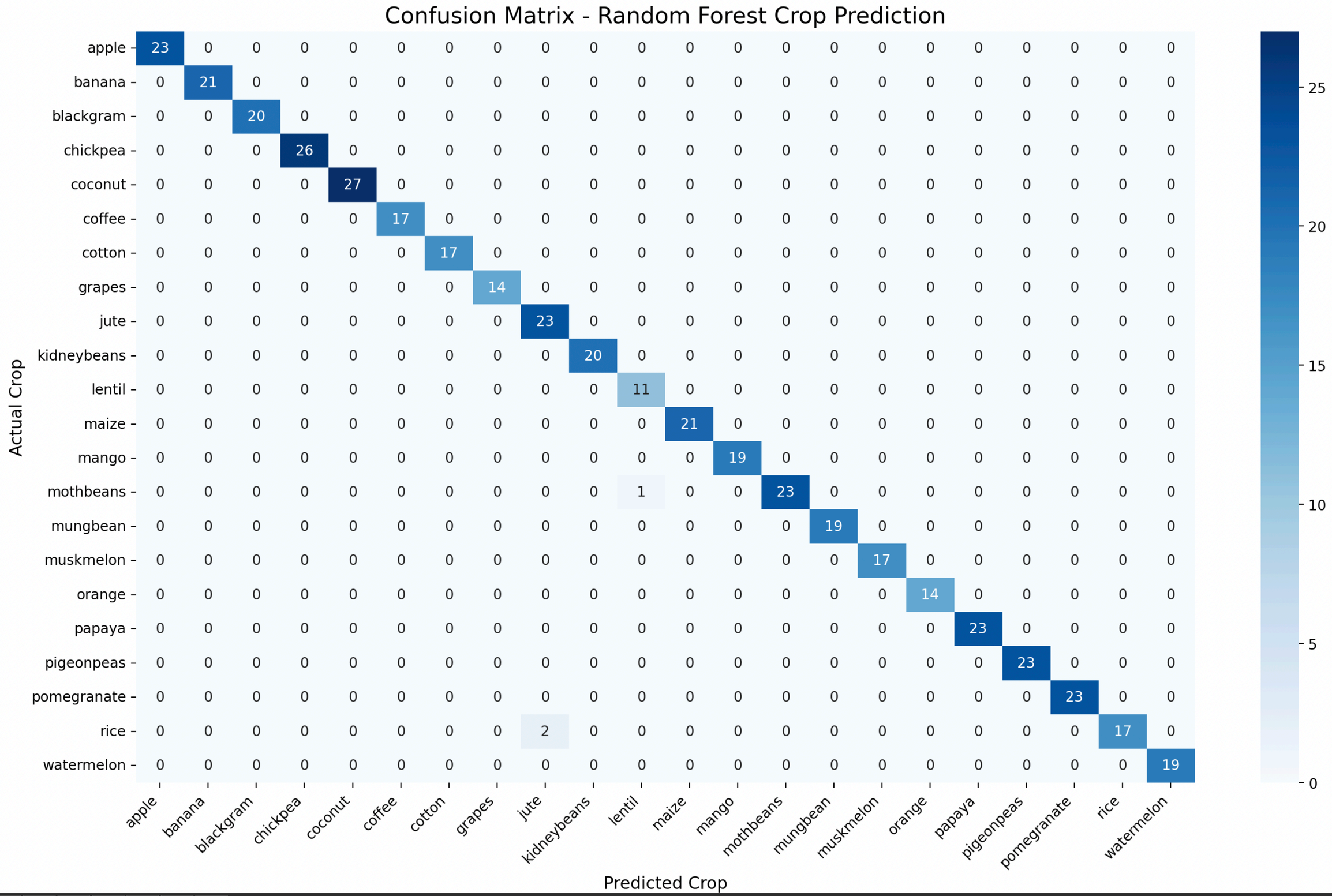
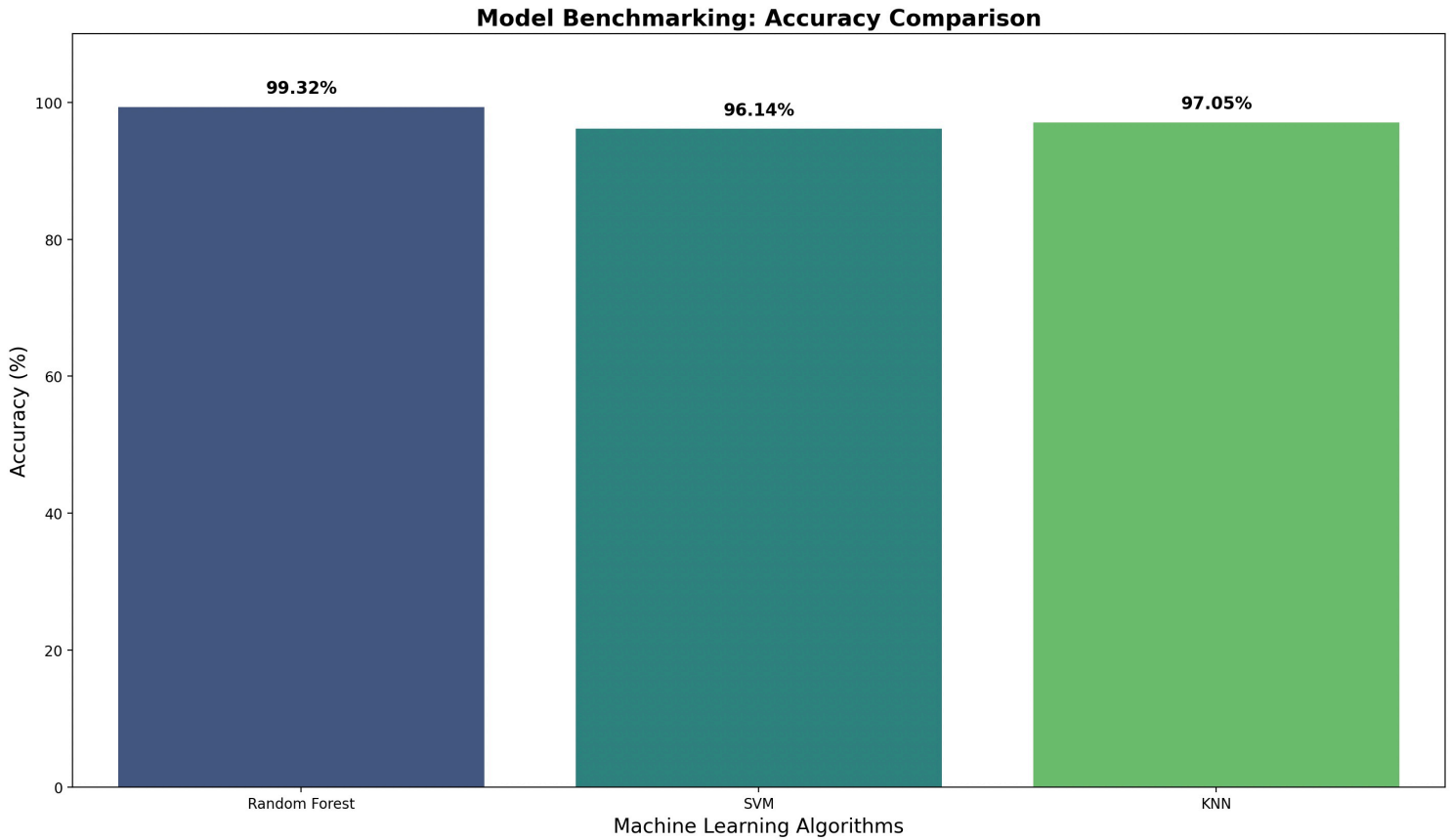

Confusion Matrix - Random Forest Crop Prediction
Actual Crop
Predicted Crop
apple
banana
blackgram
chickpea
coconut
coffee
cotton
grapes
jute
kidneybeans
lentil
maize
mango
mothbeans
mungbean
muskmelon
orange
papaya
pigeonpeas
pomegranate
rice
watermelon
25
20
15
10
5
0


Model Benchmarking: Accuracy Comparison
99.32%
96.14%
97.05%
100
80
60
40
20
0
Accuracy (%)
Random Forest
SVM
KNN
Machine Learning Algorithms